

location using the command (also see Fig. 6):

```
$ sudo mv ~/picam_init /etc/init.d/picam
```

Next make the script executable using the command (also see Fig. 7):

```
$ sudo chmod +x /etc/init.d/picam
```

To make the script start during booting, type the command given below (also see Fig. 8):

```
pi@raspberrypi ~$ sudo mv ~/picam_init /etc/init.d/picam
```

Fig. 6: Moving the script to desired location

```
pi@raspberrypi ~$ sudo chmod +x /etc/init.d/picam
```

Fig. 7: Making the script executable

```
pi@raspberrypi ~$ sudo update-rc.d picam defaults
```

Fig. 8: Making the script start at boot time

```
pi@raspberrypi ~$ sudo apt-get install motion
```

Fig. 9: Installing motion software

```
pi@raspberrypi ~$ sudo apt-get install -y libjpeg62 libjpeg62-dev libformat53 libavformat-dev libavcodec53 libavcodec-dev libavutil51 libavutil-dev libdc1394-dev libmpeg2-dev libmpeg2-4
```

Fig. 10: Installing all the dependencies

```
pi@raspberrypi ~$ tar -zxvf motion-mmalcam.tar.gz
```

Fig. 11: Extracting the folder

```
pi@raspberrypi ~$ ./motion -n -c motion-mmalcam.conf
```

Fig. 12: Running the motion application

```
$ sudo update-rc.d picam default
```

The script will now start and shut down along with the Raspberry Pi. We can also control it manually. That is, to stop the script, use the command:

```
/etc/init.d/picam stop
```

To start the script, use the command:

```
/etc/init.d/picam start
```

Method 2: Using motion software.

To install motion software, use the command below (also see Fig. 9):

```
$ sudo apt-get install motion
```

The standard motion packages do not yet work with Raspberry Pi camera. For that we have to install the special binary code known as motion-

mmal. So install following dependencies (also see Fig. 10):

```
$ sudo apt-get install -y libjpeg62 libjpeg62-dev libavformat53 libavformat-dev libavcodec53 libavcodec-dev libavutil51 libavutil-dev libdc1394-dev libmpeg2-dev libmpeg2-4 libmjpegclient-dev libjpeg5 libjpeg-dev
```

After installing all the dependencies, download motion software from the link:

```
https://www.dropbox.com/s/xdfoxcm5hu7ls97d/motion-mmal.tar.gz
```

Extract it using the command (also see Fig. 11):

```
tar -zxvf motion-mmal.tar.gz
```

This will extract the motion application as well as a configuration file. Run the extracted motion application along with its configuration file using the command (also see Fig. 12):

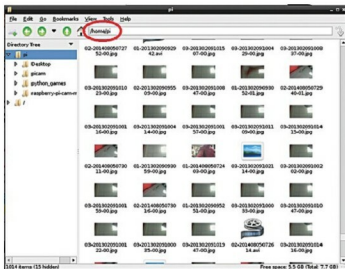


Fig. 14: Captured images stored in Raspberry Pi

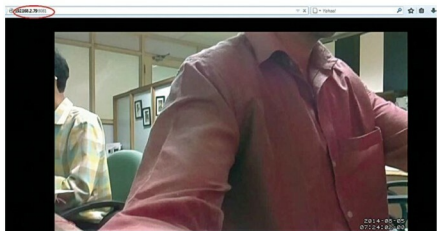


Fig. 13: Streaming video

```
./motion -n -c motion-mmalcam.conf
```

While motion application is running, open the browser and type the IP address of your Raspberry Pi with the 8081 port (for example, 192.128.2.79:8081), as shown in Fig. 13, to see the live streaming video.

When camera detects the motion it will also capture the image and store it in Raspberry Pi Home folder, as shown in Fig. 14. ●

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